ESP32 Power Wheels Controller

Professional User Manual

Can-Am Maverick X3 Conversion Project

Firmware Version: 2.1
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⚠ SAFETY WARNING

READ THIS ENTIRE MANUAL BEFORE OPERATING THE VEHICLE

This is a modified children's ride-on vehicle with significantly increased power and capabilities compared to the original. Improper use, assembly, or maintenance can result in:

- **Electrical shock** (24V system)
- **Fire hazard** (high current draw)
- **Injury** from increased speed and power
- **Property damage**

Safety Requirements:

 ALWAYS supervise children during operation
 ALWAYS wear appropriate safety gear (helmet recommended)
 NEVER operate on steep hills or near traffic
 NEVER exceed weight capacity
 NEVER operate with damaged or loose wiring
 ALWAYS disconnect battery when servicing
 ALWAYS check battery voltage before use

Adults Only: Assembly, maintenance, and electrical work should only be performed by adults with appropriate technical knowledge.

Table of Contents

1. Introduction
2. System Overview
3. Hardware Specifications
4. Component List
5. Safety Systems
6. Assembly Instructions
7. Wiring Guide
8. Software Setup
9. Operation Guide
10. Control Modes
11. Troubleshooting
12. Maintenance Schedule
13. Technical Specifications
14. Warranty & Support

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1. Introduction

1.1 Project Overview

This manual covers the ESP32-based replacement control system for the Power Wheels Can-Am Maverick X3 ride-on vehicle. The original control board has been replaced with a modern, feature-rich system offering:

- **Dual Control Modes:** Manual (pedal/shifter) and Remote (radio control)
- **Advanced Motor Control:** Independent 4-wheel drive with BTS7960 H-bridge drivers
- **Safety Features:** Voltage monitoring, overcurrent protection, thermal management
- **Radio Control:** ExpressLRS 2.4GHz long-range system
- **Real-time Telemetry:** Battery, current, and motor monitoring

1.2 Key Features

 4x Independent Motor Control - Each wheel controlled separately
 24V High-Power System - 4x 200W motors (800W total)
 Dual Control Modes - Switch between manual and radio control

 Multi-speed Operation - Reverse, Low, and Fast modes
 Advanced Safety Systems - Multiple protection mechanisms
 Real-time Monitoring - Battery voltage, current, temperature
 Radio Failsafe - Automatic shutdown if signal lost
 Expandable - I/O expander and display support

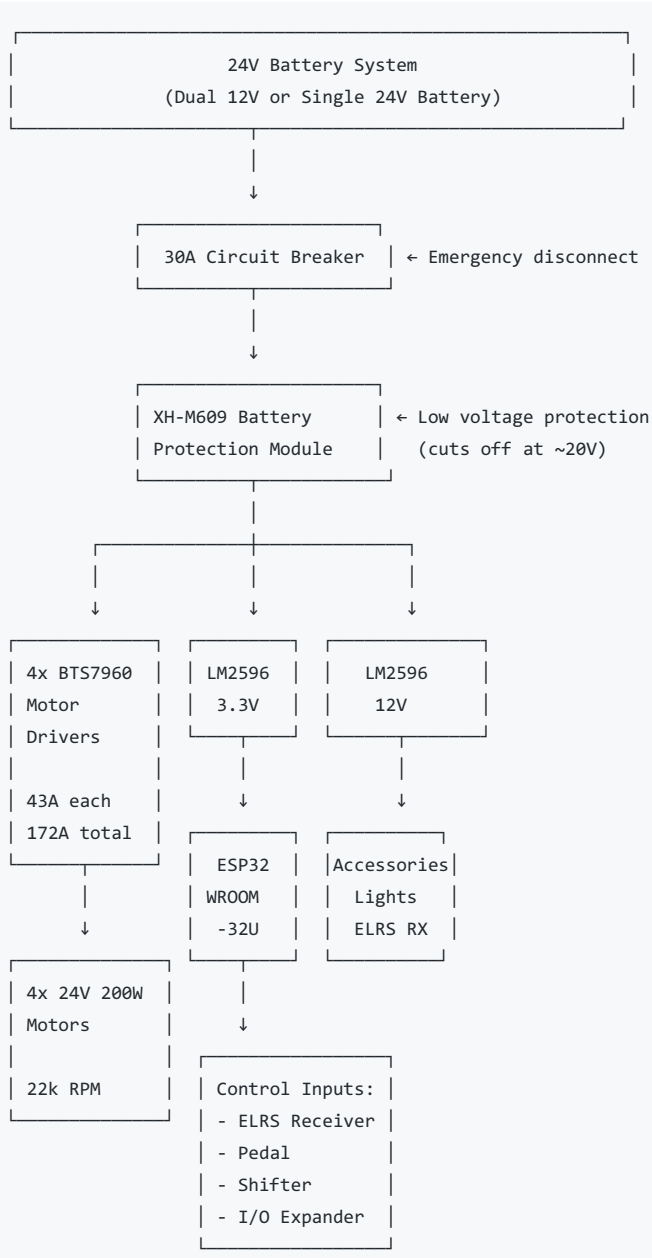
1.3 Who This Manual Is For

- **Primary Users:** Parents/guardians assembling and maintaining the vehicle
- **Secondary Users:** Advanced hobbyists and makers
- **Required Skills:** Basic electronics knowledge, soldering, multimeter use
- **Tools Needed:** See Section 6.1

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2. System Overview

2.1 System Architecture



2.2 Signal Flow

Manual Mode:

```
Pedal Input → ADC → ESP32 → PWM → BTS7960 → Motors
Shifter → GPIO → Direction Control
```

Remote Mode:

```
Transmitter → ELRS RX → CRSF Serial → ESP32 → PWM → BTS7960 → Motors
                                          ↓
                                    Failsafe Monitor
```

2.3 Power Distribution

Component	Voltage	Current	Source
Motors (4x)	24V	8-10A each (33A total)	Battery → XH-M609 → BTS7960
ESP32	3.3V	~200mA	LM2596 #1 (3.3V output)
BTS7960 Logic	3.3V	~50mA each	Shared with ESP32
ELRS Receiver	5V (rec.)	60mA	LM2596 #2 (12V) → 5V regulator
Lights	12V	Varies	LM2596 #2 (12V output)
MCP23017 Expander	3.3V	~5mA	Shared with ESP32
OLED Display	3.3V	~20mA	Shared with ESP32

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3. Hardware Specifications

3.1 Main Components

ESP32-WROOM-32U Microcontroller

- **Processor:** Dual-core Xtensa LX6 @ 240MHz
- **RAM:** 520KB SRAM
- **Flash:** 4MB
- **GPIO:** 34 pins (26 usable in this project)
- **ADC:** 12-bit, 18 channels (using ADC1 only)
- **PWM:** 16 channels (using 8 for motors)
- **Communication:** UART, I2C, SPI
- **Power:** 3.3V @ 200mA typical

BTS7960 Motor Drivers (4 units)

- **Main Chip:** Infineon BTS7960 H-Bridge
- **Continuous Current:** 43A per driver
- **Peak Current:** 100A per driver (short duration)
- **Operating Voltage:** 6V-27V (motor power)
- **Logic Voltage:** 3.3V-5V compatible
- **PWM Frequency:** Up to 25kHz (configured at 1kHz)
- **Protection:** Thermal shutdown, overcurrent, short circuit
- **Current Sensing:** R_IS and L_IS outputs (1mV/A typical)
- **Control:** Dual PWM (RPWM forward, LPWM reverse)

Motors (4 units)

- **Type:** Brushed DC motors
- **Voltage:** 24V nominal
- **Power:** 200W each (800W total)
- **Speed:** 22,000 RPM no-load
- **Current:** 8-10A typical per motor under load
- **Peak Current:** 15-20A per motor (startup/stall)

Power Supply Components

Batteries:

- Configuration: Dual 12V in series OR single 24V
- Type: Lead-acid or LiPo (6S)
- Capacity: Recommended 7Ah minimum

- Voltage Range: 19V (cutoff) to 27V (full charge)

XH-M609 Battery Protection:

- Low voltage disconnect: ~20V (adjustable)
- Over-voltage disconnect: ~29V (adjustable)
- Current capacity: 30A continuous

30A Circuit Breaker:

- Manual reset type
- Mounting location: Easily accessible
- Purpose: Emergency disconnect and overcurrent protection

LM2596 Buck Converters (2 units):

- Input: 4.5V-40V (powered from 24V)
- Output #1: 3.3V @ 3A (for ESP32 and logic)
- Output #2: 12V @ 2A (for accessories)
- Efficiency: ~92%
- Protection: Thermal shutdown, current limiting

Radio Control System

Jumper AION-RX-MINI Receiver:

- Protocol: ExpressLRS (CRSF)
- Frequency: 2.4GHz
- Channels: 16
- Range: Up to 5km
- Voltage: 3.5V-6V
- Current: 60mA @ 5V
- Output: Serial CRSF @ 420000 baud
- Latency: <10ms typical

Transmitter: Jumper T-Lite V2 (not included)

- ExpressLRS 2.4GHz compatible
- Multi-protocol support
- Telemetry capable

3.2 Optional Components

MCP23017 I/O Expander:

- Additional GPIO: 16 pins (8+8 ports)
- Interface: I2C @ 0x20
- Purpose: Individual thermal monitoring, audio control

SSD1306 OLED Display:

- Size: 0.91" (128x32 pixels)
- Interface: I2C @ 0x3C
- Purpose: Real-time telemetry display

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4. Component List

4.1 Required Components

Electronics:

- [ ] 1x ESP32-WROOM-32U development board
- [ ] 4x BTS7960 motor driver modules (43A)
- [ ] 4x 24V 200W DC motors (included with vehicle)
- [ ] 1x Jumper AION-RX-MINI ELRS receiver
- [ ] 1x Jumper T-Lite V2 transmitter (or compatible)
- [ ] 2x LM2596 buck converter modules
- [ ] 1x XH-M609 battery protection module
- [ ] 1x 30A resettable circuit breaker
- [ ] 1x 5V voltage regulator (LM7805 or AMS1117-5.0)

Passive Components:

- [ ] 2x 100kΩ resistor (1%, 1/4W) - voltage divider
- [ ] 1x 12kΩ resistor (1%, 1/4W) - voltage divider
- [ ] 4x 1N4148 diodes - thermal fault OR circuit
- [ ] 1x 10kΩ resistor (1/4W) - pull-up for thermal fault
- [ ] Assorted capacitors (0.1µF, 10µF) - noise filtering

Wiring & Connectors:

- ☐ 12 AWG wire (red/black) - 24V power lines (2 meters)
- ☐ 14 AWG wire (red/black) - motor connections (2 meters)
- ☐ 22 AWG wire (various colors) - signal lines (5 meters)
- ☐ Dupont connectors & crimp pins (assortment)
- ☐ Heat shrink tubing (various sizes)
- ☐ Wire terminals (ring, spade, bullet connectors)
- ☐ Cable ties (zip ties, various sizes)
- ☐ Terminal blocks (5-10 pieces)

Hardware & Mounting:

- ☐ Mounting board/plate (acrylic or wood)
- ☐ M3 standoffs and screws
- ☐ Double-sided tape or adhesive mounting pads
- ☐ Velcro straps (battery/component securing)

4.2 Optional Components

- ☐ 1x MCP23017 I/O expander module (TENSTAR or similar)
- ☐ 1x 0.91" SSD1306 OLED display
- ☐ 4x DS18B20 temperature sensors (for detailed thermal monitoring)
- ☐ 1x ACS712-30B current sensor (alternative to BTS7960 built-in)
- ☐ 4x Heatsinks for BTS7960 (usually included with modules)
- ☐ Cooling fans (12V, 40mm or 60mm)

4.3 Tools Required

Essential Tools:

- ☐ Soldering iron (temperature controlled, 60W minimum)
- ☐ Solder (rosin-core, 60/40 or lead-free)
- ☐ Wire strippers (for 12-22 AWG)
- ☐ Wire cutters (flush cutters recommended)
- ☐ Multimeter (voltage, current, continuity testing)
- ☐ Crimping tool (for Dupont connectors)
- ☐ Screwdrivers (Phillips, flathead, various sizes)
- ☐ Allen/hex keys (metric set)
- ☐ Heat gun or lighter (for heat shrink)

Recommended Tools:

- ☐ Helping hands or PCB holder
- ☐ Bench power supply (for testing)
- ☐ Oscilloscope (for debugging PWM signals)
- ☐ Label maker (for wire identification)
- ☐ Hot glue gun (strain relief)

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5. Safety Systems

5.1 Hardware Protection

1. Battery Protection (XH-M609)

Purpose: Prevents battery over-discharge and over-voltage
Trigger Conditions:

- Low voltage: <20V (adjustable)
- High voltage: >29V (adjustable) **Action:** Disconnects battery output **Recovery:** Manual reset or automatic when voltage returns to safe range

2. Circuit Breaker (30A)

Purpose: Overcurrent and emergency disconnect
Trigger Condition: Current draw exceeds 30A
Action: Opens circuit (manual reset required)

Location: Main battery positive line, easily accessible

3. BTS7960 Thermal Protection

Purpose: Prevents driver overheating
Trigger Condition: Internal chip temperature >150°C
Action: Automatic shutdown of affected driver
Recovery: Automatic when cooled below ~130°C

4. BTS7960 Current Limiting

Purpose: Prevents overcurrent damage
Specification: 43A continuous, 100A peak
Action: Built-in current limiting and thermal foldback

5.2 Software Protection

1. Low Voltage Cutoff

Threshold: 19.0V (configurable in code)
Warning: 21.0V
Action: Disables all motors, logs warning
Display: "SAFETY: Low battery voltage - Motors disabled!"

2. Overcurrent Protection

Limit: 40.0A total (configurable)
Warning: 35.0A
Action: Emergency stop, motors disabled
Display: "SAFETY: Overcurrent detected - Motors disabled!"

3. Thermal Monitoring

Trigger: Over-temperature fault pin (active LOW)
Recovery Time: 30 seconds forced cooldown
Action: Emergency stop, motors disabled
Display: "!!! OVER-TEMPERATURE FAULT DETECTED !!!"

4. Radio Failsafe (Remote Mode Only)

Timeout: 1000ms (1 second) no valid signal
Detection: CRSF channel values out of range (900-2100)
Action: Immediate emergency stop
Display: "SAFETY: Radio signal lost - Motors disabled!"
Recovery: Automatic when signal restored

5. Motor Ramping

Purpose: Prevents sudden torque changes (gear damage prevention)
Ramp Rate: 10 PWM units per 20ms (0-100% in ~510ms)
Application: All speed changes (acceleration and deceleration)

5.3 Safety System Priority

Highest Priority → Lowest Priority:

1. **Hardware Protection** (Circuit breaker, XH-M609) - Cannot be overridden
2. **Thermal Shutdown** (BTS7960 chip-level) - Automatic hardware protection
3. **Software Emergency Stop** (Overcurrent, over-temperature) - Immediate
4. **Radio Failsafe** (Remote mode only) - 1-second timeout
5. **Low Voltage Warning** (Pre-cutoff alert) - Graceful shutdown warning
6. **Motor Ramping** (Mechanical protection) - Always active

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6. Assembly Instructions

6.1 Pre-Assembly Checklist

Before starting assembly:

□ Read this entire manual
□ Gather all components (Section 4)
□ Prepare workspace (clean, well-lit, static-safe)
□ Charge batteries or use bench power supply for testing
□ Have multimeter ready for continuity/voltage checks
□ Disconnect ALL power sources during assembly

6.2 Component Placement

Recommended Layout:

```
Vehicle Layout (Top View):

    [Front]
┌─────────────────┐
│  [Motor FL]     │  [Motor FR]
│                 │
│   [BTS7960x4]   │ ← Mount drivers together for shared cooling
│   [ESP32]       │ ← Central location, away from motors
│   [MCP23017]    │ ← Next to ESP32
│   [OLED Display]│ ← Visible location (dashboard or rear)
│                 │
│  [LM2596 #1]    │ ← 3.3V regulator
│  [LM2596 #2]    │ ← 12V regulator
│  [XH-M609]      │ ← Battery protection
│                 │
│  [Battery Pack] │ ← Secure with Velcro
│  [Circuit Break]│ ← Accessible location
│                 │
│  [Motor RL]     │  [Motor RR]
└─────────────────┘
    [Rear]

[ELRS Receiver] → Mount on exterior with antenna pointing up
```

Critical Placement Notes:

- ⚠ Keep ESP32 away from motors (EMI protection)
- ⚠ Mount BTS7960 drivers with good airflow
- ⚠ Circuit breaker must be easily accessible
- ⚠ ELRS antenna should point vertically for best range
- ⚠ Secure batteries firmly (no movement during operation)

6.3 Assembly Steps

Step 1: Prepare Mounting Board

1. Measure vehicle electronics bay
2. Cut mounting board (acrylic/wood) to fit
3. Plan component layout (refer to diagram above)
4. Mark mounting holes for standoffs
5. Drill holes (3mm for M3 screws)
6. Test-fit all components before wiring

Step 2: Mount Components

Mount in this order (easier access for wiring):

1. **Circuit breaker** → Front, accessible
2. **XH-M609** → Near battery
3. **LM2596 modules** → Accessible for adjustment
4. **BTS7960 drivers** (4x) → Grouped together, good airflow
5. **ESP32** → Central location
6. **MCP23017** → Next to ESP32
7. **OLED display** → Visible location (decide later based on use)

Mounting Methods:

- M3 standoffs for PCBs
- Double-sided tape for modules
- Hot glue for strain relief (after testing!)

Step 3: Power Distribution Wiring

⚠ CRITICAL: Follow this order exactly to prevent shorts

3A. Battery to Circuit Breaker:

```
Battery+ (red, 12 AWG) → Circuit Breaker Input
Battery- (black, 12 AWG) → Common GND point
```

3B. Circuit Breaker to XH-M609:

```
Circuit Breaker Output → XH-M609 IN+
Common GND → XH-M609 IN-
```

3C. XH-M609 to Buck Converters:

```
XH-M609 OUT+ (red, 12 AWG) →
   ├→ LM2596 #1 Input+ (3.3V regulator)
   ├→ LM2596 #2 Input+ (12V regulator)
   └→ All 4 BTS7960 B+ terminals

XH-M609 OUT- (black, 12 AWG) →
   └→ Common GND (star ground configuration)
```

3D. Buck Converter Outputs:

```
LM2596 #1 (3.3V) →
   ├→ ESP32 VIN (3.3V)
   ├→ All BTS7960 VCC pins (logic power)
   ├→ MCP23017 VCC
   └→ OLED VCC

LM2596 #2 (12V) →
   ├→ 5V regulator input (for ELRS receiver)
   └→ Lights/accessories

5V regulator output →
   └→ ELRS receiver VCC
```

⚠ **Before Proceeding:**

1. Use multimeter to verify:
    - LM2596 #1 outputs exactly 3.3V
    - LM2596 #2 outputs exactly 12V
    - 5V regulator outputs exactly 5.0V
2. Check for shorts between power and ground
3. Disconnect power after verification

Step 4: Motor Driver Signal Wiring

For Each BTS7960 Module (repeat 4 times):

Front-Left Motor (BTS7960 #1):

```
BTS7960 Pin → ESP32 Pin
├─ RPWM → GPIO 25
├─ LPWM → GPIO 26
├─ R_EN → GPIO 21 (shared with all 4 drivers)
├─ L_EN → GPIO 21 (shared with all 4 drivers)
├─ VCC → 3.3V (from LM2596 #1)
├─ GND → Common GND
├─ R_IS → GPIO 39 (optional - conflicts with shifter in manual mode)
└─ L_IS → GPIO 36 (optional)

Motor Connections:
├─ B+ → 24V from XH-M609
├─ B- → Common GND
├─ M+ → Motor positive (red wire)
└─ M- → Motor negative (black wire)
```

Front-Right Motor (BTS7960 #2):

```
BTS7960 Pin → ESP32 Pin
├─ RPWM → GPIO 27
├─ LPWM → GPIO 14
├─ R_EN → GPIO 21 (shared)
├─ L_EN → GPIO 21 (shared)
├─ VCC → 3.3V
├─ GND → Common GND
├─ R_IS → GPIO 34 (optional - conflicts with pedal in manual mode)
└─ L_IS → (not connected - GPIO 35 used for voltage monitoring)
```

Rear-Left Motor (BTS7960 #3):

```
BTS7960 Pin → ESP32 Pin
├─ RPWM → GPIO 12 (⚠ Note: boot strapping pin)
├─ LPWM → GPIO 13
├─ R_EN → GPIO 21 (shared)
├─ L_EN → GPIO 21 (shared)
├─ VCC → 3.3V
├─ GND → Common GND
├─ R_IS → (not connected - GPIO conflict)
└─ L_IS → (not connected - GPIO conflict)
```

Rear-Right Motor (BTS7960 #4):

```
BTS7960 Pin → ESP32 Pin
├─ RPWM → GPIO 32
├─ LPWM → GPIO 33
├─ R_EN → GPIO 21 (shared)
├─ L_EN → GPIO 21 (shared)
├─ VCC → 3.3V
├─ GND → Common GND
├─ R_IS → (not connected - GPIO conflict)
└─ L_IS → (not connected - GPIO conflict)
```

Motor Enable (Shared):

```
ESP32 GPIO 21 →
   ├─→ BTS7960 #1 R_EN + L_EN
   ├─→ BTS7960 #2 R_EN + L_EN
   ├─→ BTS7960 #3 R_EN + L_EN
   └─→ BTS7960 #4 R_EN + L_EN
```

Step 5: ELRS Receiver Wiring

```
AION-RX-MINI → ESP32
├─ GND → Common GND
├─ VCC → 5V from 5V regulator
├─ TX (CRSF) → GPIO 16 (ESP32 Serial2 RX)
└─ RX → GPIO 17 (ESP32 Serial2 TX - optional telemetry)
```

Antenna Placement:

- Mount receiver on exterior of vehicle
- Antenna should point vertically upward
- Keep antenna away from metal and motors
- Secure with double-sided tape or Velcro

Step 6: Manual Control Inputs

Pedal (Potentiometer):

```
Pedal Potentiometer:
├─ Terminal 1 → 3.3V
├─ Wiper (middle) → GPIO 34 (⚠ Shared with current sense)
└─ Terminal 3 → GND
```

Shifter (2-position switches):

```
Shifter Drive Position:
├─ One side → GPIO 39 (INPUT_PULLUP)
└─ Other side → GND

Shifter Reverse Position:
├─ One side → GPIO 5 (INPUT_PULLUP)
└─ Other side → GND
```

Note: When both switches open (not pressed) = PARK mode

Step 7: Telemetry & Monitoring

Battery Voltage Divider:

```
24V Battery+ ─┬─[100kΩ resistor]─┬─[12kΩ resistor]─ GND
              │                 │
              │                 └─→ GPIO 35 (ADC)
              │
             [Add 0.1µF capacitor to GND for noise filtering]
```

⚠ **CRITICAL VERIFICATION:**

- Use multimeter to measure voltage at GPIO 35
- At 24V battery: Should read ~2.57V at GPIO 35
- At 27V battery: Should read ~2.89V at GPIO 35
- **MUST BE BELOW 3.3V** or ESP32 will be damaged!

Thermal Fault Monitoring (Combined):

```
BTS7960 #1 FAULT ──[1N4148 diode, cathode→]──┐
BTS7960 #2 FAULT ──[1N4148 diode, cathode→]──┤
BTS7960 #3 FAULT ──[1N4148 diode, cathode→]──┼── GPIO 19
BTS7960 #4 FAULT ──[1N4148 diode, cathode→]──┘      │
                                                 [10kΩ]
                                                    │
                                                  3.3V
All FAULT grounds → Common GND
```

Lights Control:

```
ESP32 GPIO 22 → Relay/MOSFET gate → 12V Lights
```

Step 8: I/O Expander (Optional - MCP23017)

```
MCP23017 → ESP32
├─ VCC → 3.3V
├─ GND → Common GND
├─ SDA → GPIO 23
├─ SCL → GPIO 18
├─ A0 → GND (address: 0x20)
├─ A1 → GND
└─ A2 → GND

MCP23017 GPIOs (if using individual thermal monitoring):
├─ GPA0 → BTS7960 #1 FAULT
├─ GPA1 → BTS7960 #2 FAULT
├─ GPA2 → BTS7960 #3 FAULT
├─ GPA3 → BTS7960 #4 FAULT
└─ GPA4-GPB7 → Spare for future use
```

Step 9: OLED Display (Optional)

```
SSD1306 OLED → ESP32
├─ VCC → 3.3V
├─ GND → Common GND
├─ SDA → GPIO 23 (shared with MCP23017)
└─ SCL → GPIO 18 (shared with MCP23017)
```

Note: OLED and MCP23017 can share the same I2C bus (different addresses)

6.4 Post-Assembly Checks

Before powering on, verify:

☐ All power connections tight and insulated
☐ No exposed wires or bare conductors
☐ Correct polarity on all connections
☐ LM2596 outputs set to correct voltages (3.3V and 12V)
☐ Voltage divider verified with multimeter (<3.3V at GPIO 35)
☐ Motor wires not touching each other
☐ ELRS antenna not obstructed
☐ All grounds connected to common ground
☐ Circuit breaker in accessible location
☐ Batteries secured and cannot move

---

7. Wiring Guide

7.1 Wire Gauge Selection

Connection Type	Wire Gauge	Max Current	Color Code
Battery main	12 AWG	30A+	Red (positive), Black (negative)
Motor power	14 AWG	20A	Red (positive), Black (negative)
BTS7960 B+/B-	14 AWG	20A	Red (positive), Black (negative)
ESP32 logic	22-24 AWG	<1A	Various colors (see below)
Signal lines	22-26 AWG	<100mA	Various colors

7.2 Color Coding Standard

Power:

- Red: Positive (+)

- Black: Negative (-) / Ground
- Yellow: 3.3V logic
- Orange: 5V logic
- Blue: 12V

Signals:

- Green: PWM signals
- White: Digital inputs
- Brown: Analog inputs
- Purple: I2C (SDA/SCL)
- Gray: UART (TX/RX)

7.3 Connector Types

Connection	Recommended Connector
Battery	XT60 or Anderson Powerpole
Motors	Bullet connectors (4mm) or screw terminals
BTS7960 control	Dupont 2.54mm headers
ESP32	Dupont 2.54mm headers or soldered
Power distribution	Terminal blocks (5mm pitch)

7.4 Grounding Strategy

Star Ground Configuration:

All grounds should connect to a single common point (star ground) to minimize ground loops:



DO NOT create ground loops!

---

8. Software Setup

8.1 Prerequisites

Required Software:

- PlatformIO IDE (VSCode extension) OR Arduino IDE 2.x
- USB cable (USB-A to Micro-USB or USB-C, depending on ESP32 board)
- ESP32 board support package
- Libraries (see Section 8.3)

8.2 IDE Installation

Option 1: PlatformIO (Recommended)

1. Install Visual Studio Code from https://code.visualstudio.com/
2. Open VSCode → Extensions (Ctrl+Shift+X)
3. Search for "PlatformIO IDE"
4. Click Install
5. Restart VSCode

Option 2: Arduino IDE

1. Download Arduino IDE 2.x from https://www.arduino.cc/en/software
2. Install and open Arduino IDE
3. Go to File → Preferences
4. Add ESP32 board manager URL:

   ```
   https://espressif.github.io/arduino-esp32/package_esp32_index.json
   ```

5. Go to Tools → Board → Boards Manager
6. Search for "ESP32"
7. Install "esp32 by Espressif Systems" (version 3.x or later)

8.3 Libraries Required

Install these libraries:

1. **AlfredoCRSF** - CRSF protocol for ELRS receiver
   - GitHub: https://github.com/AlfredoSystems/AlfredoCRSF
2. **Adafruit MCP23X17** (if using I/O expander)
   - Library Manager: "Adafruit MCP23017 Arduino Library"
3. **Adafruit SSD1306** (if using OLED display)
   - Library Manager: "Adafruit SSD1306"
4. **Adafruit GFX** (required for SSD1306)
   - Library Manager: "Adafruit GFX Library"

PlatformIO: Already configured in `platformio.ini`

Arduino IDE:

- Go to Tools → Manage Libraries
- Search and install each library

8.4 Firmware Upload

Step 1: Connect ESP32

1. Connect ESP32 to computer via USB cable
2. Wait for drivers to install (Windows may need manual driver installation)
3. Note the COM port (Windows) or /dev/ttyUSB0 (Linux) or /dev/cu.* (Mac)

Step 2: Configure Board Settings

PlatformIO: Already configured in `platformio.ini`

Arduino IDE:

- Tools → Board → ESP32 Dev Module
- Upload Speed: 921600
- CPU Frequency: 240MHz
- Flash Frequency: 80MHz
- Flash Mode: QIO
- Flash Size: 4MB
- Partition Scheme: Default 4MB with spiffs
- Port: Select your ESP32's COM port

Step 3: Compile & Upload

PlatformIO:

1. Open project folder in VSCode
2. Click PlatformIO icon (alien head) in left sidebar
3. Click "Upload" under env:esp32dev

Arduino IDE:

1. Open `sketch_jun22b.ino`
2. Click Verify (✓) to compile
3. Click Upload (→) to flash firmware

Step 4: Verify Upload

1. Open Serial Monitor (115200 baud)
2. Press ESP32 reset button
3. You should see:

```
[100 ms] ESP32 Power Wheels Controller Starting...
[200 ms] CRSF receiver initialized on Serial2
[300 ms] BTS7960 motor drivers enabled
[500 ms] Setup complete. System ready.
```

8.5 Configuration Options

In `sketch_jun22b.ino`:

Development Mode (Line ~287):

```
#define DEVELOPMENT_MODE false  // Set true for bench testing without hardware
```

Safety Thresholds (Lines ~220-235):

```
const float LOW_VOLTAGE_CUTOFF = 19.0;   // Adjust for your battery type
const float LOW_VOLTAGE_WARNING = 21.0;
const float MAX_CURRENT_LIMIT = 40.0;    // Maximum safe current
const float HIGH_CURRENT_WARNING = 35.0;
```

Radio Failsafe (Line ~238):

```
const unsigned long RADIO_TIMEOUT_MS = 1000;  // 1 second timeout
```

Motor Ramping (Lines ~241-243):

```
const int RAMP_STEP = 10;       // Increase for faster acceleration
const int RAMP_DELAY_MS = 20;   // Decrease for faster response
```

Current Sensitivity (Line ~149):

```
const float BTS7960_CURRENT_SENSITIVITY = 0.001;  // 1mV/A (adjust per your module)
```

Voltage Divider Ratio (Line ~213):

```
const float VOLTAGE_DIVIDER_RATIO = 9.333;  // For 100k/12k divider
```

---

9. Operation Guide

9.1 Pre-Operation Checklist

Before each use:

□ Check battery voltage (>21V for safe operation)
□ Inspect all wiring for damage or loose connections
□ Verify motors spin freely (no mechanical binding)
□ Test circuit breaker (accessible and functional)
□ Check tire pressure and condition
□ Verify weight capacity not exceeded
□ Ensure operating area is safe (flat, clear of obstacles)
□ Bind ELRS receiver to transmitter (if not already bound)

9.2 Startup Procedure

1. **Power On Sequence:**
    - Close circuit breaker (if open)
    - Wait for ESP32 to boot (~2 seconds)
    - Listen for initialization sounds (optional buzzer)
    - Check Serial Monitor output (if connected)
2. **Verify Initialization:**
    - ELRS receiver LED should be solid (bound) or flashing (searching)
    - OLED display (if installed) shows startup screen
    - No error beeps or warnings
3. **Radio Check (Remote Mode):**

- Turn on transmitter
- Wait for receiver to connect (solid LED)
- Move sticks - verify response (check telemetry or display)
- Test failsafe: turn off transmitter → vehicle should stop

4. **Manual Mode Check:**
    - Set mode switch to Manual
    - Press pedal slightly → no movement (should be in PARK)
    - Move shifter to DRIVE → verify direction indicator
    - Press pedal → slow movement
    - Move shifter to REVERSE → verify direction change

9.3 Control Modes

Manual Mode (Pedal & Shifter Control)

Shifter Positions:

- **PARK:** Both switches open → No motor movement (safe state)
- **DRIVE:** D switch closed → Forward direction enabled
- **REVERSE:** R switch closed → Reverse direction enabled

Pedal Operation:

- No press: 0% speed
- Light press: ~30% speed
- Medium press: ~60% speed
- Full press: 100% speed

Lights: Always ON in manual mode

Speed Limiting: None (full power available in manual mode)

Remote Mode (Radio Control)

Required Setup:

- Transmitter: Jumper T-Lite V2 or compatible ELRS radio
- Receiver: Must be bound and connected (solid LED)
- Mode switch: Set to REMOTE

Channel Mapping:

Channel	Control	Function	Values
0	Throttle	Left stick vertical	1000 (idle) → 2000 (full)
1	Steering	Right stick horizontal	1000 (left) → 2000 (right)
2	Horn	Button/switch	>1500 = active
3	Mode Switch	2-position	<1500 Manual, >1500 Remote
5	Lights	2-position	<1500 Off, >1500 On
6	Speed Mode	3-position	<1300 Rev, 1300-1700 Low, >1700 Fast

Speed Modes:

1. **REVERSE (<1300):**
    - Motors run in reverse
    - Full power available
    - Use caution - same power as forward
2. **FORWARD LOW (1300-1700):**
    - Forward direction
    - Speed limited to 50% maximum
    - Best for beginners or tight spaces
    - Safer for younger children
3. **FORWARD FAST (>1700):**
    - Forward direction
    - Full power available
    - Maximum speed
    - For experienced drivers only

Failsafe Behavior:

- Radio signal lost for >1 second
- All motors immediately stop
- Vehicle enters safe state
- Warning logged (if display installed)
- Automatic recovery when signal restored

9.4 Speed Control

Acceleration:

- Gradual ramping prevents gear damage
- 0 → 100% in ~510ms
- Adjust RAMP_STEP in code for faster/slower response

Deceleration:

- Same ramping applies when reducing speed
- Emergency stop bypasses ramping (instant)

Braking:

- Release throttle/pedal → gradual deceleration
- Motors electrically brake (not coast)

9.5 Emergency Procedures

Emergency Stop Methods:

1. **Circuit Breaker:**
    - Location: Front of vehicle (accessible)
    - Action: Flip breaker switch to OFF
    - Result: Immediate power disconnect
    - Use for: Electrical emergency, runaway vehicle
2. **Release Controls:**
    - Manual: Release pedal
    - Remote: Release throttle stick to center
    - Result: Gradual deceleration (~500ms)
    - Use for: Normal stops, controlled slowdown
3. **Mode Switch:**
    - Switch to Manual mode and shifter to PARK
    - Result: Immediate motor disable
    - Use for: Quick stop when in remote mode
4. **Turn Off Transmitter (Remote Mode):**
    - Turn off transmitter
    - Result: Failsafe triggers, motors stop within 1 second
    - Use for: Testing failsafe, emergency in remote mode

Common Issues During Operation:

Vehicle stops suddenly:

- Check battery voltage (may be <19V)
- Check for over-temperature (let cool 30 seconds)
- Check radio signal (if in remote mode)
- Check serial monitor for error messages

Vehicle sluggish:

- Low battery voltage (recharge)
- Motors overheating (let cool down)
- Mechanical binding (check wheels/gears)
- Check motor current (may be overloaded)

Erratic behavior:

- Radio interference (move away from WiFi routers)
- Loose wiring (inspect connections)
- Low transmitter battery (replace)
- Check serial monitor for diagnostic info

9.6 Shutdown Procedure

1. **Stop Vehicle:**
    - Bring to complete stop (release throttle/pedal)
    - Shifter to PARK (manual mode)
    - Turn off transmitter (remote mode)
2. **Power Off:**

    - Open circuit breaker
    - Wait 5 seconds for capacitors to discharge

3. **Post-Operation Checks:**
    - Feel BTS7960 heatsinks (should not be too hot to touch)
    - Check battery voltage (note for next charge cycle)
    - Inspect wiring for any damage or looseness
    - Check tires and mechanical components

---

10. Control Modes

10.1 Manual Mode Details

Purpose: Traditional Power Wheels operation using physical controls

Best For:

- Younger children (3-6 years)
- Learning to drive
- Close supervision
- Predictable operation

Characteristics:

- Physical pedal provides tactile feedback
- Shifter gives clear indication of direction
- Lights always on (visibility)
- No radio dependence
- Simple operation

Limitations:

- No speed limiting (full power in both directions)
- No failsafe (relies on physical controls only)
- Current sensing limited (pedal uses ADC pin)

10.2 Remote Mode Details

Purpose: Adult-controlled operation via radio transmitter

Best For:

- Very young children (2-4 years)
- Teaching steering skills
- Crowded areas (precise control)
- Demo/show operation
- Safety-critical situations

Characteristics:

- Complete control from transmitter
- Adjustable speed modes (Low/Fast)
- Radio failsafe protection
- Can override child's actions
- Lights controllable

Advanced Features:

- Speed mode selection (3 positions)
- Steering trim adjustment (transmitter-side)
- Telemetry feedback (optional)
- Multiple child profiles (transmitter presets)

10.3 Mode Switching

How to Switch:

- Manual → Remote: Move mode switch on transmitter to Remote position
- Remote → Manual: Move mode switch to Manual position

When Mode Changes:

- Motors immediately respond to new control source
- Shifter state read (manual mode)
- Speed mode applied (remote mode)
- Lights behavior changes

Safety During Switching:

- Current motor speed is maintained momentarily
- Ramping still applies (no sudden changes)
- If controls conflict, system chooses safe default (slower speed)

11. Troubleshooting

11.1 Power Issues

Problem: ESP32 won't power on

Symptoms: No LED, no serial output, no activity

Possible Causes:

1. No 3.3V from LM2596 #1
2. Loose/broken USB cable (if powering via USB)
3. Damaged ESP32

Solutions:

- Measure voltage at LM2596 #1 output (should be 3.3V)
- Adjust LM2596 potentiometer if needed
- Try different USB cable
- Check for shorts between 3.3V and GND
- Verify ESP32 VIN pin connection

Problem: BTS7960 drivers not working

Symptoms: Motors don't run, no PWM output

Possible Causes:

1. No 24V at B+ terminal
2. R_EN/L_EN not enabled
3. BTS7960 in thermal shutdown
4. Damaged driver

Solutions:

- Measure voltage at B+ (should be 24V)
- Check GPIO 21 is HIGH (motor enable)
- Feel heatsink (if very hot, wait for cooldown)
- Test with different motor
- Swap BTS7960 with known working unit

Problem: System reboots randomly

Symptoms: ESP32 resets during operation

Possible Causes:

1. Brownout (voltage dip below 2.8V)
2. Overcurrent on 3.3V rail
3. EMI from motors
4. Loose power connection

Solutions:

- Add bulk capacitor (1000µF) on 3.3V rail
- Check all 3.3V connections are tight
- Route signal wires away from motor wires
- Add ferrite beads on motor power lines
- Verify LM2596 output voltage stable under load

11.2 Motor Issues

Problem: Motors don't spin

Symptoms: No movement when controls applied

Possible Causes:

1. Safety system active (check serial monitor)
2. Motors not connected
3. BTS7960 not enabled
4. Mechanical binding

Solutions:

- Check serial monitor for error messages:
  - Low voltage? → Charge battery
  - Overcurrent? → Check for short circuit
  - Over-temp? → Wait 30 seconds for cooldown
- Verify motor connections (M+ and M-)
- Check GPIO 21 (motor enable) is HIGH
- Manually spin wheels (should turn freely)

- Test motors with bench power supply

Problem: Motors spin wrong direction

Symptoms: Forward command causes reverse (or vice versa)

Solutions:

1. Swap M+ and M- wires on that motor
2. OR change pin assignments in code:

```
// Swap RPWM and LPWM pin numbers for that motor
```

Problem: One motor weaker than others

Symptoms: Vehicle pulls to one side, uneven current draw

Possible Causes:

1. Damaged motor
2. Mechanical binding on that wheel
3. Loose motor connection
4. Weak BTS7960 driver

Solutions:

- Check motor current (serial telemetry)
- Swap motors to isolate problem
- Inspect gearbox for damage
- Verify all motor connections tight
- Test with different BTS7960 driver

Problem: Motors stutter or jerk

Symptoms: Unsmooth operation, pulsing

Possible Causes:

1. Low battery voltage
2. Loose connections
3. PWM frequency too low
4. EMI interference

Solutions:

- Charge battery (check voltage >21V)
- Tighten all power connections
- Increase PWM frequency in code (try 5kHz):

```
const int PWM_FREQ = 5000;
```

- Add capacitors (0.1µF) across motor terminals

11.3 Radio Issues

Problem: ELRS receiver won't bind

Symptoms: LED flashing rapidly, no connection

Solutions:

1. Put transmitter in bind mode first
2. Power on receiver (should flash green)
3. Wait 20 seconds for WiFi update mode
4. Access receiver via WiFi (http://10.0.0.1)
5. Manually bind via web interface
6. Verify correct ELRS firmware version (2.4 GHz)

Problem: Frequent failsafes

Symptoms: Vehicle stops randomly, signal lost messages

Possible Causes:

1. Low transmitter battery
2. Antenna damaged or poorly positioned
3. EMI from motors
4. Out of range

Solutions:

- Replace transmitter batteries
- Reposition receiver antenna (vertical, clear view)

- Move receiver away from BTS7960 drivers
- Reduce distance to vehicle
- Check transmitter output power setting

Problem: High latency

Symptoms: Delayed response to stick inputs

Possible Causes:

1. ELRS mode set to long-range (slower update rate)
2. Radio interference

Solutions:

- Change ELRS mode to faster rate (e.g., 250Hz)
- Move away from WiFi routers
- Check for other 2.4GHz devices

11.4 Sensor Issues

Problem: Battery voltage reading incorrect

Symptoms: Voltage much higher/lower than actual battery voltage

Possible Causes:

1. Wrong voltage divider ratio in code
2. Resistor values incorrect
3. Loose connection

Solutions:

- Measure actual battery voltage with multimeter
- Measure voltage at GPIO 35 (should be battery / 9.333)
- Verify resistor values (100kΩ and 12kΩ)
- Adjust VOLTAGE_DIVIDER_RATIO in code if needed:

```
// Measured values:
// Battery = 24.0V
// GPIO 35 = 2.5V
// Ratio = 24.0 / 2.5 = 9.6
const float VOLTAGE_DIVIDER_RATIO = 9.6;
```

Problem: Current reading always zero

Symptoms: Motor current shows 0.0A even when motors running

Possible Causes:

1. R_IS/L_IS pins not connected
2. Wrong sensitivity value
3. Reading in manual mode (GPIO conflict)

Solutions:

- Verify R_IS/L_IS wiring to ESP32 ADC pins
- Check current sense only works in REMOTE mode
- Try different sensitivity value:

```
const float BTS7960_CURRENT_SENSITIVITY = 0.0005; // Try 0.5mV/A
```

- Measure voltage at R_IS pin with multimeter (should be >0V when motor running)

Problem: Thermal fault triggering incorrectly

Symptoms: Over-temp warning when drivers are cool

Possible Causes:

1. Diode orientation wrong
2. Pull-up resistor missing
3. Loose connection
4. GPIO 19 floating

Solutions:

- Check diode orientation (cathode toward GPIO 19)
- Verify 10kΩ pull-up resistor from GPIO 19 to 3.3V
- Test: disconnect fault pins, check if problem persists
- Short GPIO 19 to GND → should trigger fault (if not, wiring issue)

11.5 Software Issues

Problem: Firmware won't compile

Error Examples:

- `ledcAttach() not found`
- `Library not found`
- `Undefined reference`

Solutions:

- **For ledcAttach() error:** Use ESP32 board package v3.x or later
  - Arduino IDE: Update via Boards Manager
  - PlatformIO: Check `platformio.ini` platform version
- **For library errors:** Install missing libraries (see Section 8.3)
- **Check serial output** for specific error details

Problem: Firmware won't upload

Error Examples:

- `Failed to connect to ESP32`
- `A fatal error occurred`

Solutions:

1. Hold BOOT button on ESP32 while clicking upload
2. Try different USB cable
3. Try different USB port
4. Install/update CH340 or CP2102 drivers
5. Reduce upload speed:
   - Arduino IDE: Tools → Upload Speed → 115200
   - PlatformIO: Set `upload_speed = 115200` in platformio.ini

Problem: Serial monitor shows gibberish

Symptoms: Random characters, unreadable output

Solutions:

- Set baud rate to 115200 in serial monitor
- Press ESP32 reset button after opening monitor
- Check USB cable quality

---

12. Maintenance Schedule

12.1 Before Each Use

□ Visual inspection of all wiring
□ Check battery voltage (>21V)
□ Test circuit breaker function
□ Verify motors spin freely
□ Check tire pressure
□ Test radio connection (if using remote mode)

12.2 Weekly (or every 5 hours of operation)

□ Clean dust/debris from vehicle
□ Tighten any loose connections
□ Check BTS7960 heatsink attachment
□ Inspect motor brushes (if accessible)
□ Lubricate moving parts (wheels, steering)
□ Check battery terminals for corrosion

12.3 Monthly (or every 20 hours of operation)

□ Full system inspection
□ Test all safety systems:

- Low voltage cutoff (discharge battery to 19V)
- Overcurrent protection (load test)
- Thermal shutdown (heat driver with heat gun - carefully!)
- Radio failsafe (turn off transmitter) □ Clean battery terminals
  □ Check motor current balance (all 4 should be similar)
  □ Verify voltage divider accuracy
  □ Update firmware if new version available
  □ Backup configuration settings

12.4 Seasonal (or every 50 hours of operation)

- Deep clean all components
- Replace thermal paste on BTS7960 heatsinks
- Inspect motor brushes, replace if worn
- Check gearbox lubrication
- Test battery capacity (discharge under load)
- Inspect all solder joints
- Re-apply heat shrink tubing if damaged
- Check ESP32 for signs of overheating

12.5 Battery Maintenance

Lead-Acid Batteries:

- Charge after each use
- Don't store discharged (<50%)
- Check water level monthly (if not sealed)
- Replace every 1-2 years

LiPo Batteries (6S):

- Charge after each use (don't leave discharged)
- Store at storage voltage (3.8V per cell = 22.8V)
- Balance charge every 10 cycles
- Inspect for puffing/damage before each use
- Replace immediately if damaged
- Never discharge below 18V (3.0V per cell)

Charging Safety:

- Use appropriate charger for battery type
- Charge in fireproof location
- Never leave charging unattended
- Disconnect if battery gets hot
- Follow manufacturer's charging current limits

---

13. Technical Specifications

13.1 Electrical Specifications

Parameter	Specification
System Voltage	24V nominal (19V-27V operating range)
Total Power	800W (4x 200W motors)
Maximum Current	40A (33A typical, 40A limit)
Peak Current	60A (startup, limited by circuit breaker to 30A)
Logic Voltage	3.3V
PWM Frequency	1 kHz (configurable 100Hz-25kHz)
PWM Resolution	8-bit (0-255)
ADC Resolution	12-bit (0-4095)
Control Loop Rate	~100Hz (10ms cycle time)

13.2 Performance Specifications

Parameter	Value
Top Speed	~5-8 mph (depends on gearing, load)
Acceleration	0-max in ~2 seconds (with ramping)
Braking Distance	~3-5 feet from max speed

Parameter	Value
Runtime	30-60 minutes (depends on battery capacity, usage)
Weight Capacity	Up to 130 lbs (59 kg) - check original vehicle specs
Operating Temperature	0°C to 40°C ambient

13.3 Radio Specifications

Parameter	Specification
Protocol	ExpressLRS (CRSF)
Frequency	2.4 GHz
Channels	16 (6 used)
Update Rate	50Hz-500Hz (mode dependent)
Latency	<10ms typical
Range	Up to 5km (clear line of sight)
Typical Range	300-500m practical use
Failsafe Timeout	1 second

13.4 Protection Thresholds

Protection	Threshold	Action
Low Voltage Cutoff	<19.0V	Emergency stop
Low Voltage Warning	<21.0V	Warning message
Overcurrent Limit	>40.0A	Emergency stop
Overcurrent Warning	>35.0A	Warning message
Thermal Shutdown	BTS7960 internal (~150°C)	Emergency stop
Radio Failsafe	>1000ms no signal	Emergency stop
Current Imbalance	>5A deviation between motors	Warning only

13.5 Dimensions & Weight

Component	Dimensions	Weight
ESP32 Module	55 x 28 mm	10g
BTS7960 (each)	51 x 41 mm + heatsink	60g
MCP23017	38 x 18 mm	5g
OLED Display	38 x 13 mm	3g
LM2596	43 x 21 mm	15g
Total Electronics	Varies (mounting dependent)	~500g

14. Warranty & Support

14.1 Disclaimer

This is a DIY project based on open-source hardware and software. There is **NO WARRANTY** expressed or implied. Use at your own risk.

The builder accepts all responsibility for:

- Safe assembly and operation
- Compliance with local laws and regulations
- Supervision of children using the vehicle
- Proper maintenance and inspection
- Any damage or injury resulting from use

14.2 Safety Certifications

This project does **NOT** have any safety certifications (UL, CE, FCC, etc.). It is intended for:

- **Personal use only**
- **Non-commercial applications**
- **Educational purposes**
- **Adult supervision required**

14.3 Support Resources

Documentation:

- This user manual
- Code comments in `sketch_jun22b.ino`
- Additional guides in `docs/` folder:
  - BTS7960_WIRING_GUIDE.md
  - BTS7960_TELEMETRY_GUIDE.md
  - GPIO_PIN_ALLOCATION.md
  - THERMAL_MONITORING_OPTIONS.md
  - CHANGELOG.md

Online Resources:

- ESP32 documentation: https://docs.espressif.com/
- AlfredoCRSF library: https://github.com/AlfredoSystems/AlfredoCRSF
- ExpressLRS: https://www.expresslrs.org/
- Arduino forums: https://forum.arduino.cc/

Component Datasheets:

- ESP32-WROOM-32U datasheet
- BTS7960 datasheet (Infineon)
- MCP23017 datasheet (Microchip)
- SSD1306 datasheet (Solomon Systech)

14.4 Community

Share your build, ask questions, and help others:

- Reddit: r/esp32, r/arduino, r/rccars
- Facebook: Power Wheels modification groups
- YouTube: Search "ESP32 power wheels"

14.5 Contributing

Found an issue or have an improvement?

- Report bugs via GitHub issues (if project is hosted)
- Submit improvements via pull requests
- Share your modifications and enhancements
- Help others in forums and communities

---

15. Appendices

Appendix A: Bill of Materials (BOM)

See Section 4 for complete component list with quantities.

Appendix B: Schematic Diagram

Refer to `docs/WIRING_GUIDE.md` for detailed wiring diagrams.

Appendix C: Pinout Reference

See `docs/GPIO_PIN_ALLOCATION.md` for complete pin usage table.

Appendix D: Firmware Configuration Reference

Key configuration constants in `sketch_jun22b.ino`:

```
// Motor Control
const int PWM_FREQ = 1000;        // PWM frequency (Hz)
const int PWM_RES = 8;            // PWM resolution (bits)
const int RAMP_STEP = 10;         // Ramping increment
const int RAMP_DELAY_MS = 20;     // Ramping delay

// Safety Thresholds
const float LOW_VOLTAGE_CUTOFF = 19.0;     // Volts
const float LOW_VOLTAGE_WARNING = 21.0;    // Volts
const float MAX_CURRENT_LIMIT = 40.0;      // Amps
const float HIGH_CURRENT_WARNING = 35.0;   // Amps
const unsigned long RADIO_TIMEOUT_MS = 1000; // Milliseconds
const unsigned long TEMP_RECOVERY_MS = 30000; // Milliseconds

// Calibration
const float VOLTAGE_DIVIDER_RATIO = 9.333;
const float BTS7960_CURRENT_SENSITIVITY = 0.001; // V/A
const float ADC_VREF = 3.3;       // Volts
const int ADC_MAX_VALUE = 4095;   // 12-bit

// CRSF Channels
#define THROTTLE_CH 0
#define STEERING_CH 1
#define HORN_CH 2
#define MODE_SWITCH_CH 3
#define LIGHTS_CH 5
#define SPEEDMODE_CH 6
```

Appendix E: Troubleshooting Quick Reference

Symptom	Check This First
No power	Battery voltage, circuit breaker, LM2596 outputs
Motors don't run	Serial monitor errors, GPIO 21 enable, BTS7960 connections
Wrong direction	Swap M+ and M-, or swap RPWM/LPWM in code
Random stops	Battery voltage, thermal, radio signal
Radio won't bind	Transmitter in bind mode first, correct ELRS version
Voltage wrong	Measure with multimeter, adjust divider ratio
Current always 0A	Only works in REMOTE mode, check R_IS/L_IS wiring

Appendix F: Upgrade Paths

Future Enhancement Ideas:

- Add GPS module for speed/distance tracking
- Implement data logging to SD card
- Add Bluetooth app control (smartphone interface)
- Install cooling fans on BTS7960 drivers
- Add horn/sound effects via DFPlayer Mini
- Implement cruise control (constant speed mode)
- Add headlights/taillights with automatic control
- Install suspension monitoring (accelerometer)
- Add battery capacity meter (fuel gauge IC)

- Implement regenerative braking (requires different motor drivers)

Appendix G: Legal Considerations

Important Legal Notes:

- Check local laws regarding modified children's vehicles
- Some jurisdictions restrict speed or require safety equipment
- May affect any original manufacturer warranty
- Insurance implications (personal liability)
- Age restrictions may apply
- Use on private property only (unless otherwise permitted)

Recommended Safety Equipment:

- Helmet (strongly recommended)
- Knee/elbow pads
- Closed-toe shoes
- Adult supervision at all times

Document History

Version	Date	Changes
1.0	December 2025	Initial release

Contact & Feedback

For questions, suggestions, or to share your build:

- **Email:** [Your contact info]
- **GitHub:** [Your repository URL]
- **Forum:** [Your support forum]

END OF USER MANUAL

Thank you for building this project! Drive safely and have fun!

Document prepared for: ESP32 Power Wheels Controller Project
Firmware Version 2.1
Can-Am Maverick X3 Conversion

Document Type: Professional User Manual
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